

2025

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: **CACS252** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What do you mean by true error? Find a positive root of equation $(x^2 = 1)$, which lies between 0 and 1 using bisection method.
3. What is interpolation? Given the following set of data points, obtained the table of divided difference. Use the table to estimate the value of $f(1.5)$.

x_i	0	1	2	3	4
$f(x_i)$	0.1234	0.2345	0.3456	0.4567	0.5678
4. Solve the following system of equations by Gauss elimination method.
$$\begin{cases} 2x + y + 4z = 12 \\ 8x - 3y + 2z = 23 \\ 4x + 11y - z = 33 \end{cases}$$
5. The distance of motorbike at interval 2 minutes are given below:

Time(minutes)	0	2	4	6	8
Distance(Km)	0	1.53	3.86	6.7	10.2

Evaluate the velocity and acceleration of motorbike at time 2.37 and 4.57 minutes.
6. Write a program to compute value of y when $x = 2$ such that $\frac{dy}{dx} = 3x^2 + 1$ with $y(1) = 2$ and $h = 0.25$ using Euler's method.
7. Find the largest Eigen value and corresponding Eigen vector of the following matrix using power method.
$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$
8. Solve the Poisson equation $\nabla^2 f = 2x^2y^2$ over the square domain $(0 \leq x \leq 3)$ and $(0 \leq y \leq 3)$ with $f = 0$ on the boundary and $h = 1$.

Group C

Attempt any TWO questions[2x10=20]

9. What do you mean by ill condition? Solve the following system using Jacobi iteration method.
$$\begin{cases} 2a + b + c = 5 \\ 3a + 5b + 2c = 15 \\ 2a + b + 4c = 8 \end{cases}$$
10. How interpolation is differ from regression? Write down algorithm and program for Lagrange interpolation polynomial.
11. What is fixed point iteration method? How it converge the root of non-linear equation? Find the root of equation $f(x) = x^2 - 3x + 2$ using N-R method.

